

Mathematics for Computing

Assignment

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Outline

- 1 Sets
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- 3 Types of Sets
- 4 Set Operations
- 5 Inclusion–Exclusion Principle
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Sets

- A **set** is a well-defined collection of distinct objects, called **elements** or **members**.
- Sets are usually denoted by capital letters ($A, B, C, \dots$) and elements by lowercase letters.
- Membership is written as $x \in A$ (x belongs to A) or $x \notin A$ (x does not belong to A).
- A set can be represented in two main ways:
 - ▶ **Roster (Tabular) Form:** listing all elements, e.g. $A = \{1, 2, 3, 4, 5\}$
 - ▶ **Set Builder (Rule) Form:** describing a property satisfied by the elements

Set Builder Form

- In set builder form, a set is written as:

$$A = \{ x \mid P(x) \}$$

read as “ A is the set of all x such that x satisfies property P ”.

- It is useful when listing all elements is impossible or impractical (e.g. infinite sets).

Set Builder Form — 10 Examples (1/2)

① $A = \{x \mid x \in \mathbb{N}, x < 10\} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

② $B = \{x \mid x \in \mathbb{Z}, x^2 = 4\} = \{-2, 2\}$

③ $C = \{x \mid x \text{ is a vowel in the English alphabet}\} = \{a, e, i, o, u\}$

④ $D = \{x \mid x \in \mathbb{R}, 0 \leq x \leq 1\}$ (the closed interval $[0, 1]$)

⑤ $E = \{x \mid x \in \mathbb{N}, x \text{ is even}, x \leq 20\}$

Set Builder Form — 10 Examples (2/2)

- ⑥ $F = \{x \mid x \in \mathbb{N}, x \text{ is prime, } x < 30\}$
- ⑦ $G = \{x \mid x \in \mathbb{N}, x = n^2 \text{ for some } n \in \mathbb{N}, x < 100\}$
- ⑧ $H = \{x \mid x = 2n, n \in \mathbb{N}\}$ (all positive multiples of 2)
- ⑨ $I = \{x \mid x \text{ is a day of the week starting with the letter } S\} = \{\text{Saturday, Sunday}\}$
- ⑩ $J = \{(x, y) \mid x, y \in \mathbb{R}, x^2 + y^2 = 1\}$ (points on the unit circle)

Types of Sets — AI/DS Examples (1/2)

- **Empty (Null) Set** $\emptyset$: has no elements.
E.g. $\{x \mid x \text{ is a test sample with negative age}\} = \emptyset$
- **Singleton Set**: exactly one element.
E.g. $\{\text{accuracy}\}$ — a model evaluated using a single metric
- **Finite Set**: countable number of elements.
E.g. $\{\text{cat, dog, bird}\}$ — class labels in a classification problem
- **Infinite Set**: unending number of elements.
E.g. the set of all possible real-valued weights in a neural network layer
- **Equal Sets** ($A = B$): same elements, regardless of order.
E.g. the set of unique tokens in two identical text documents
- **Equivalent Sets** ($A \sim B$): same cardinality, not necessarily same elements.
E.g. $\{5 \text{ input features}\} \sim \{5 \text{ hyperparameters}\}$

Types of Sets — AI/DS Examples (2/2)

- **Subset** ($A \subseteq B$): every element of A is in B .
E.g. selected features $\subseteq$ all available features
- **Proper Subset** ($A \subset B$): $A \subseteq B$ and $A \neq B$.
E.g. training set $\subset$ full dataset
- **Universal Set** U : the set containing all objects under consideration.
E.g. $U = \{0, 1, \dots, 255\}$ — all possible pixel intensities in an 8-bit image
- **Power Set** $\mathcal{P}(A)$: the set of all subsets of A ; $|\mathcal{P}(A)| = 2^{|A|}$.
E.g. $\mathcal{P}(\text{feature set})$ — all possible feature subsets tried during feature selection
- **Disjoint Sets** ($A \cap B = \emptyset$): sets with no common elements.
E.g. the training set and the test set are disjoint (no overlapping samples)

Set Operations — Definitions and a Simple Example

Let $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, and universal set $U = \{1, 2, 3, 4, 5\}$.

- **Union:** $A \cup B = \{x \mid x \in A \text{ or } x \in B\} = \{1, 2, 3, 4\}$
- **Intersection:** $A \cap B = \{x \mid x \in A \text{ and } x \in B\} = \{2, 3\}$
- **Difference:** $A - B = \{x \mid x \in A \text{ and } x \notin B\} = \{1\}$
- **Complement:** $A' = \{x \mid x \in U \text{ and } x \notin A\} = \{4, 5\}$
- **Symmetric Difference:** $A \triangle B = (A - B) \cup (B - A) = \{1, 4\}$

Inclusion–Exclusion Principle — Three Sets

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

Inclusion–Exclusion Principle — General (n Sets)

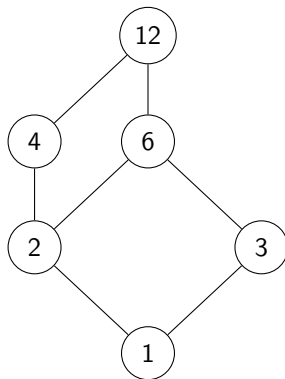
$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{i=1}^n |A_i| - \sum_{i < j} |A_i \cap A_j| + \sum_{i < j < k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} \left| \bigcap_{i=1}^n A_i \right|$$

Hasse Diagrams

- A **Hasse diagram** is a graphical representation of a finite **partially ordered set (poset)**.
- Rules for drawing:
 - ▶ Loops (self-relations) are omitted.
 - ▶ Edges implied by transitivity are omitted.
 - ▶ If $a \prec b$ (a is related to b , $a \neq b$), b is drawn **above** a , connected by a line (no arrowhead needed).
- Commonly used to represent divisibility, subset relations, etc.

Hasse Diagram — Example

Divisibility relation on $A = \{1, 2, 3, 4, 6, 12\}$



$a \prec b$ here means a divides b . Note that the edge 1–12 is omitted since it follows by transitivity.

Thank You